ETL Pipeline Exercise: Extracting Data

Explore how to extract and consolidate social media data from production databases using Python and SQL in an ETL pipeline. Understand incremental data loading to efficiently transfer recent records and prepare them for analysis in a data warehouse.

A case study

Suppose we're data engineers working for a digital company and we're tasked with creating an ETL pipeline.

Our company, "Fakebook," has created a social media application that users use worldwide. This application constantly generates data stored in the company's production database for management.

The company wants to process and analyze the data collected by the application to generate insights and identify usage patterns. However, these analyses in the production database will introduce a heavy load. This is why the company has decided to separate the computing and storage of the data and perform all the analysis in a separate repository called the data warehouse.

Because of that, we're tasked with creating and scheduling an ETL pipeline to transfer social media related data from our company's production database to the data warehouse.

Ask

According to the business requirements, the ETL pipeline should be scheduled to run every four hours and use incremental loading methods to load the latest data

generated by the application.

Extract

The first task is to acquire the relevant data. According to our scenario, we need to extract the latest data from the production database. For this example, the production database comprises three tables: users, posts, and comments.

The users table stores information about social media users.

Users Table

Column	Type	Descriptio
username	VARCHAR(50)	Username of the
email	VARCHAR(100)	Email address of t
date	DATE	Date when the use
followers	INT	Number of follo
location	VARCHAR(100)	User's locati

The posts table stores social media posts made by users.

Posts Table

Column	Type	Description
post_id	INT (PK)	Unique identifie
user_id	INT (PK)	Foreign key to link with u

Ask

post_text	TEXT	Text content of the pos
post_date	DATETIME	Date and time when the post v

The comments table stores comments made on social media posts.

Comments Table

Column	Type	Description
comment_id	INT (PK)	Unique identifier for the co
post_id	INT (PK)	Foreign key to link with p
user_id	INT (FK)	Foreign key to link with u
comment_text	TEXT	Text content of the comr
comment_date	DATETIME	Date and time when the comr made

Our goal is to extract and merge relevant data from these tables according to the schema of the data warehouse. Let's check what the facts table in the data warehouse looks like. This table stores the fact information about comments and posts made on social media posts:

Facts Table

Column	Type	D
user_id	INT (FK)	Foreign key referencin
post_id	INT (FK)	Foreign key referencin

Ask

comment_id	INT (FK)	Foreign key referencing c
username	VARCHAR(50)	Username of th
email	VARCHAR(100)	Email address of
followers	INT	Number of follo
post_text	TEXT	Text content of t
post_date	TIMESTAMP	Date and time when the po
comment_text	TEXT	Text content of the
comment_date	TIMESTAMP	Date and time when the co
followers_level	VARCHAR(30)	low:1000-, medium:1 high:10000

To complete the extraction step of the pipeline, we need to connect to the database using Python, write a SQL query that will merge the three tables, and filter the appropriate columns according to the fact table in the data warehouse. Finally, we save the script in a file called `helper.py`.

Remember that we're using incremental loading, so before each extraction, we need to collect the date of the most recent data in the fact table in the data warehouse.

Once we have the latest date from the data warehouse, we can extract all the rows inserted after it from the production database.

> **Note**: Fill in the missing code snippets.

Find the latest date

Ask

First, connect to the data warehouse using Python and create a SQL query to find out the date of the most recent data in the data warehouse. We'll use this query in

the next task.

grab_latest_date.py Python

```python
import psycopg2

# Connect to the database
conn = psycopg2.connect(host='localhost', port = 5432, database='postgres

# Create a cursor
cur = conn.cursor()

# Execute a SQL query
cur.execute('''
    select
        comment_date
    from fact_table
```

Run

Finding the latest date in the data warehouse

Extract data

Next, merge the three tables from the production database, and choose the columns to match the fact table schema in the data warehouse. Filter the results to get only the rows inserted after the latest date in the previous task. Finally, export the extracted data to a CSV file named `social_media_data.csv`.

grab_latest_date.py Python

```python
import psycopg2
import csv
import pandas as pd

def extract():
    """
    Extract data from the database and export it to a csv file
    """
```

Ask

```
    # Connect to the database
    conn = psycopg2.connect(host='localhost', port = 5432, database='post
```

Run

Merging the tables: comments, posts and users

Here are the solution codes for the tasks:

Solution code: Find the latest date

grab_latest_date.py Python

```
import psycopg2

def get_latest_date():
    try:
        # Connect to the data warehouse
        conn = psycopg2.connect(host='localhost', port=5432, database='po
        cur = conn.cursor()

        # Execute SQL queries to get the latest dates
        cur.execute('''
            SELECT
                GREATEST(MAX(post_date), MAX(comment_date)) AS latest_dat
            FROM fact_table;
```

Run

Solution code: Finding the latest date

- **Line 6**: We connect to a PostgreSQL database.
- **Line 7**: We create a cursor object.
- **Lines 10–14**: We query for extracting the latest date in the data warehouse.
- **Line 17**: We fetch query results.

Ask

Solution code: Extract data

grab_latest_date.py Python

```python
import psycopg2
import csv
import pandas as pd


def extract():
    """
    Extract data from the database and export it to a csv file
    """

    # Get the latest date from the fact table
    latest_date = get_latest_date()
```

Solution code: Extracting data

- **Lines 17–38**: We query for merging and extracting relevant data from the production database.
- **Lines 45–48**: We export query results to a CSV file named `social_media_data.csv`.

Ask